

Unit 6.5: Cubic & Cube Root Functions
Translations Practice

Name: **ANSWER KEY**

Period: _____

Process: (1) Find h and k (2) Plot center point (h, k) (3) Shift anchor points (4) Draw the curve

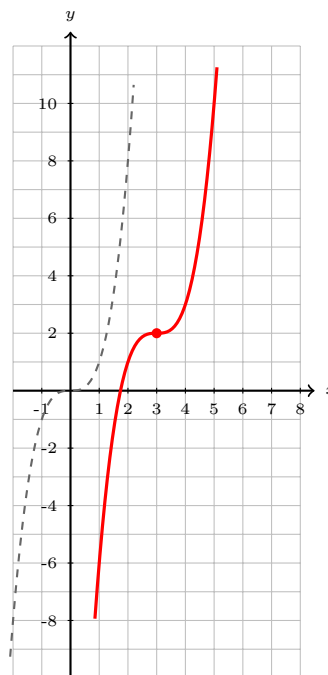
1. $f(x) = (x - 3)^3 + 2$

Verbal Desc: **Right 3, Up 2**

$h = \mathbf{3}$ $k = \mathbf{2}$

Center point: **(3, 2)**

Parent	Shifted
$(-2, -8)$	(1, -6)
$(-1, -1)$	(2, 1)
$(0, 0)$	(3, 2)
$(1, 1)$	(4, 3)
$(2, 8)$	(5, 10)



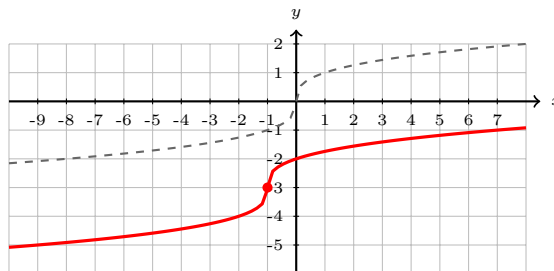
2. $f(x) = \sqrt[3]{x + 1} - 3$

Verbal Desc: **Left 1, Down 3**

$h = \mathbf{-1}$ $k = \mathbf{-3}$

Center point: **(-1, -3)**

Parent	Shifted
$(-8, -2)$	(-9, -5)
$(-1, -1)$	(-2, -4)
$(0, 0)$	(-1, -3)
$(1, 1)$	(0, -2)
$(8, 2)$	(7, -1)



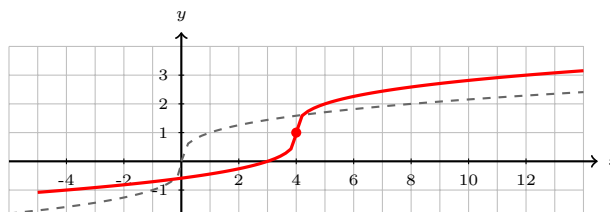
3. $f(x) = \sqrt[3]{x - 4} + 1$

Verbal Desc: **Right 4, Up 1**

$h = \mathbf{4}$ $k = \mathbf{1}$

Center point: **(4, 1)**

Parent	Shifted
$(-8, -2)$	(-4, -1)
$(-1, -1)$	(3, 0)
$(0, 0)$	(4, 1)
$(1, 1)$	(5, 2)
$(8, 2)$	(12, 3)



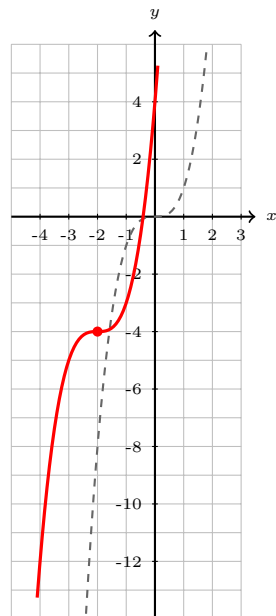
4. $f(x) = (x + 2)^3 - 4$

Verbal Desc: **Left 2, Down 4**

$h = -2$ $k = -4$

Center point: **$(-2, -4)$**

Parent	Shifted
$(-2, -8)$	$(-4, -12)$
$(-1, -1)$	$(-3, -5)$
$(0, 0)$	$(-2, -4)$
$(1, 1)$	$(-1, -3)$
$(2, 8)$	$(0, 4)$



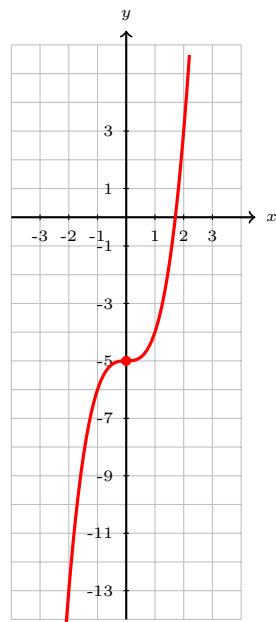
5. $f(x) = x^3 - 5$

Verbal Desc: **Down 5**

$h = 0$ $k = -5$

Center point: **$(0, -5)$**

Parent	Shifted
$(-2, -8)$	$(-2, -13)$
$(-1, -1)$	$(-1, -6)$
$(0, 0)$	$(0, -5)$
$(1, 1)$	$(1, -4)$
$(2, 8)$	$(2, 3)$



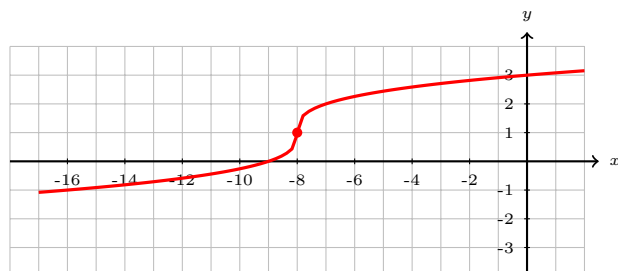
6. $f(x) = \sqrt[3]{x + 8} + 1$

Verbal Desc: **Left 8, Up 1**

$h = -8$ $k = 1$

Center point: **$(-8, 1)$**

Parent	Shifted
$(-8, -2)$	$(-16, -1)$
$(-1, -1)$	$(-9, 0)$
$(0, 0)$	$(-8, 1)$
$(1, 1)$	$(-7, 2)$
$(8, 2)$	$(0, 3)$



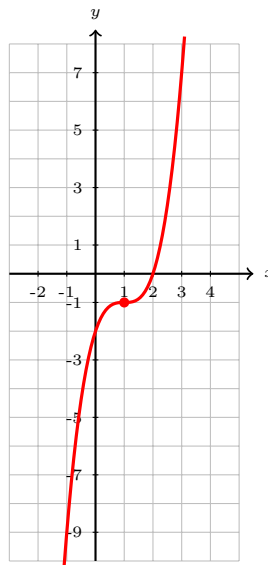
7. $f(x) = (x - 1)^3 - 1$

Verbal Desc: **Right 1, Down 1**

$h = 1 \quad k = -1$

Center point: **(1, -1)**

Parent	Shifted
$(-2, -8)$	$(-1, -9)$
$(-1, -1)$	$(0, -2)$
$(0, 0)$	$(1, -1)$
$(1, 1)$	$(2, 0)$
$(2, 8)$	$(3, 7)$

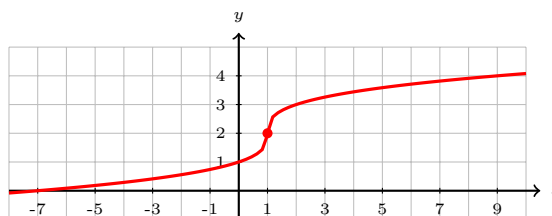


8. $f(x) = \sqrt[3]{x - 1} + 2$

$h = 1 \quad k = 2$

Verbal Desc: **Right 1, Up 2**Center point: **(1, 2)**

Parent	Shifted
$(-8, -2)$	$(-7, 0)$
$(-1, -1)$	$(0, 1)$
$(0, 0)$	$(1, 2)$
$(1, 1)$	$(2, 3)$
$(8, 2)$	$(9, 4)$

**Summary Questions:**9. Write the equation of a cubic function with center point at $(-3, 5)$.

$f(x) = (x + 3)^3 + 5$

10. Write the equation of a cube root function with center point at $(4, -2)$.

$f(x) = \sqrt[3]{x - 4} - 2$

11. A student says that $f(x) = (x + 5)^3$ shifts the graph *right* 5 units. Are they correct? Explain.**No. Inside the function (with x), the shift is opposite. $(x + 5)$ shifts LEFT 5 units.**